Sahyadri College of Arts Science & Commerce
LIBRARY
Devanahalli • Bangalore

PG – 854

I Semester M.C.A. (2 Years Course) Degree Examination, August/September 2021
(CBCS)
(2020-2021 and Onwards)
COMPUTER SCIENCE
1MCA5 : Object Oriented Programming

Time : 3 Hours | Max. Marks : 70

Instruction : *Answer* ***any five*** *from Section –* ***A*** *and* ***any four*** *from Section –* ***B****.*

SECTION – A

Answer **any 5** questions. **Each** question carries **six** marks : **(5×6=30)**

1. Define encapsulation and inheritance. Explain the purpose of 'this' and 'super' keyword in Java with example.
2. Explain different looping structures in Java.
3. Write a note on abstract method and abstract class.
4. What is a package ? Explain user defined package with example.
5. Explain final variable, final method and final class with example.
6. Write a Java program to read and write data to a file.
7. Write a note on Applet Life Cycle.
8. Explain five methods of graphics class with example.

SECTION – B

Answer **any 4** questions. **Each** question carries **10** marks : **(4×10=40)**

9. Explain salient features of Java language. **10**
10. What is a constructor ? Explain constructor overloading with an example program. **10**

P.T.O.

11. Define different types of inheritance. Explain how Java achieves multiple inheritance with an example. 10

12. Explain life cycle of a thread with a neat diagram. 10

13. What is an exception ? Explain how exception handling is achieved in Java. 10

14. Write a note on :

    a) Static variable and static methods. 5

    b) Stream classes. 5

JP – 189

I Semester M.C.A. (Two Years Course) Examination, July 2022
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
1MCA 5 : Object Oriented Programming

Time : 3 Hours Max. Marks : 70

Instructions : *1) Answer **any five** questions from Section **A**.*

*2) Answer **any four** questions from Section **B**.*

SECTION – A

Answer **any 5** questions. **Each** question carries **6** marks.

1. Explain features of Java.
2. What are different categories of operators in Java ?
3. Define inheritance. Explain the types of inheritance supported by Java.
4. What is interface ? Explain with an example how a class implements an interface.
5. What is a packages ? Explain how to create it.
6. Explain exception handling in java with an example.
7. Explain the life cycle of a thread.
8. Write a short note on graphics class.

SECTION – B

Answer **any 4** questions. **Each** question carries **10** marks.

9. a) Explain the break and continue statements with example.
   b) Explain different looping structure in Java. **(5+5)**
10. a) Explain array declaration and accessing data elements using an example.
    b) What are constructors ? Explain with an example. **(5+5)**

P.T.O.

11. a) Write a note on static variables and static methods.
    b) Explain the steps of executing an applet using a simple code. **(5+5)**

12. a) Explain user defined exception in Java.
    b) Write short note on string buffer class. **(5+5)**

13. Explain with an example the implementation of multithreading by extending thread class. **10**

14. Write a note on : **10**
    a) Synchronization
    b) Deadlock
    c) Garbage collection
    d) The switch statement.

Dr. Vyshnava Sagar College of Arts Science & Commerce • Bangalore • LIBRARY

JP – 767

I Semester M.C.A. (Two Years Course) Examination, July 2023
(CBCS Scheme)
(2020 – 2021 and Onwards)
COMPUTER SCIENCE
1MCA 5 : Object Oriented Programming

Time : 3 Hours | Max. Marks : 70

Instructions : *1) Part – **A** : Answer **any five** questions.*
*2) Part – **B** : Answer **any four** questions.*

PART – A

Answer **any five** questions. **Each** carries **six** marks. **(5×6=30)**

1. Give an account of the principles of object oriented programming.
2. What are the salient features of JAVA language ?
3. Java is rich in operators. Justify.
4. What are the roles played by precedence and associativity of operators ? Give their significance.
5. What are classes and objects ?
6. Inheritance helps in hierarchical growth of a program. Justify.
7. Why are constructors used ? What are their types ?
8. What is multithreading ? Give its importance.

Library DSCASC

PART – B

Answer **any four** questions. **Each** carries **10** marks. **(4×10=40)**

9. a) What are relational expressions and compound conditional expressions ? How are they used to change the flow of control in a program ? **5**

   b) Develop a program to accept day, month and year numbers and report whether they form a valid date or not. **5**

10. Present the syntax of the looping structures in Java and develop a Java program to display only the prime numbers in the given range of integers between m and n ($m \leq n$). **10**

11. Write a Java program to develop a class namely Book with member data, title, author, no. of pages and price. Create a list of books details and perform the following :

    a) Display all the books details.

    b) Update a books price.

    c) Sort the books details in the increasing order of price. **10**

12. Develop a program to illustrate the usage of the thread class to create multiple threads and also the usage of Runnable interface to create multiple threads. **10**

13. Develop a Java program to create a file consisting of characters and also to copy it to another file. **10**

14. Write short notes on : **(2.5×4=10)**

    a) Garbage collection.

    b) Method overriding.

    c) Abstract class.

    d) Arrays in Java.

Library DSCASC

Sagar College of Arts Science & Commerce
LIBRARY
Bangalore

PG – 370

I Semester M.C.A. (Two Years Course) Examination, June/July 2024
(CBCS) (2020-2021 and Onwards)
COMPUTER SCIENCE
1MCA5 : Object Oriented Programming

Time : 3 Hours Max. Marks : 70

Instruction : *Answer* ***any 5*** *from Section –* ***A*** *and* ***any 4*** *from Section –* ***B.***

SECTION – A

Answer **any five** questions. **Each** carries **six** marks. **(5×6=30)**

1. Explain the principles of object oriented programming.
2. Explain the categories of operators in Java.
3. Write a note on static methods and static variables.
4. What is abstract class ? Explain with example.
5. What is a package ? Explain in detail how to create and use user-defined package.
6. What is the significance of exception handling ? Explain the keywords associated with exception handling.
7. What is AWT ? Explain the methods of graphics class.
8. What are the stream classes for reading and writing into files ?

DSCASC Library

SECTION – B

Answer **any 4** questions. **Each** question carries **10** marks. **(4×10=40)**

9. Discuss in detail about
   i) Extending the thread class
   ii) Thread synchronization.
10. Write a program for accepting a string from user and perform length, concatenation, substring and replacement operations on the string using methods of string class.

P.T.O.

11. Write a note on (give example)
    a) Method overriding 5
    b) Method overloading. 5

12. What is inheritance ? List and explain types of inheritance supported by Java.

13. a) What are applets ? Write a program for creating an applet. 5
    b) Write a program to copy contents of one file to another. 5

14. Write a short note on :
    a) Static import 5
    b) Advantages and limitations of Object Oriented Programming. 5

DSCASC Library

DSCASC LIBRARY

PG – 666

I Semester M.C.A. (Two Years Course) Degree Examination, May/June 2025
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
1MCA5 : Object Oriented Programming

Time : 3 Hours | Max. Marks : 70

Instructions : *1) Answer the questions from **both** Parts.*
*2) Answer **any 5** questions from Part – **A**.*
*3) Answer **any 4** questions from Part – **B**.*

PART – A

Answer **any 5** questions. **Each** question carries **6** marks. **(5×6=30)**

1. What are the paradigms of OOP ? Explain the concept of encapsulation and inheritance with example.
2. Explain final keyword and all its uses with examples.
3. Explain object slicing, object type casting with examples.
4. Write a program to read and write a file using the Java stream classes.
5. What is an array ? Explain the arrays in Java. Write a program for array of objects.
6. Write in detail about interface. Explain how to implement interface, extend interface using syntax and example.
7. What is a thread ? Explain methods of thread class.
8. What is an applet ? Explain its life cycle with example.

DSCASC - LIBRARY

PART – B

Answer **any 4** questions. **Each** question carries **10** marks. **(4×10=40)**

9. What are the buzz words of Java ?
10. Write a Java program to illustrate static member data and static member methods.

P.T.O.

11. Write a Java program to search for a string in a list of strings and also provide code segment to sort a list of strings.

12. a) What are local classes and anonymous classes ? **5**

    b) Write a note on AWT and Graphics class. **5**

13. What is JVM ? Explain JDK, JRE in detail.

14. What is an abstract class ? Explain with an example. Also differentiate abstract classes with interfaces.

DSCASC - LIBRARY